

$$v_0 = \sqrt{2gs_0}$$

$$s = v_0 t + \frac{gt^2}{2}$$

$$2s = 2t\sqrt{2gs_0} + gt^2$$

$$g^2 t^4 - 4gt^2 s + 4s^2 = 8t^2 g s_0$$

$$g^2 t^4 + g(-4t^2 s - 8t^2 s_0) + 4s^2 = 0$$

$$g_{1,2} = \frac{4t^2 s + 8t^2 s_0 \pm \sqrt{(4t^2 s + 8t^2 s_0)^2 - 4 \cdot 4s^2 t^4}}{2t^4}$$

$$(4t^2 s + 8t^2 s_0)^2 - 16s^2 t^4 =$$

$$= \cancel{16t^4 s^2} + 64t^4 s s_0 + 64t^4 s_0^2 - \cancel{16s^2 t^4} =$$

$$= 64t^4 (s s_0 + s_0^2)$$

$$g_{1,2} = \frac{4t^2 s + 8t^2 s_0 \pm 8t^2 \sqrt{s s_0 + s_0^2}}{2t^4} =$$

$$= \frac{2s + 4s_0 \pm 4\sqrt{s s_0 + s_0^2}}{t^2}$$

$$s_0 = 9,0 \text{ cm}$$

$$s = 31,9 \text{ cm}$$

